



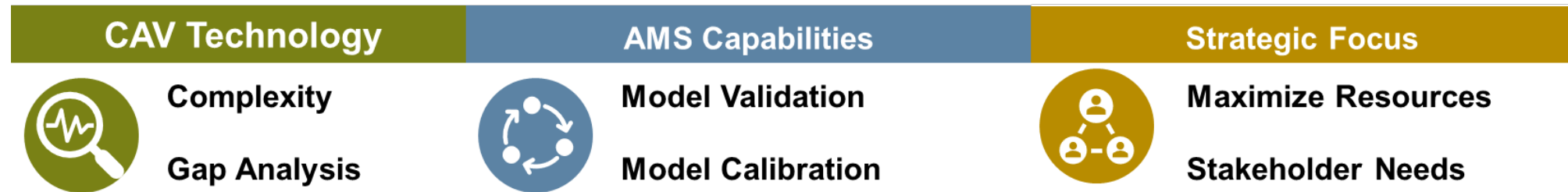
CAV AMS Program Update and CAV Technology Tracking Inventory

**ITS AMS CoP Stakeholder
Engagement**

04/22/2026

FHWA's CAV AMS Research Program Vision

- Improve AMS of ITS, including connected and automated vehicles (CAVs).



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Background Information

- **FHWA CAV AMS Research Program Objective:** To support State and local agency decision making by providing improved tools to understand the most likely impacts of Advanced Driver Assistance Systems (ADAS), Automated Driving Systems (ADS), and Cooperative Driving Automation (CDA), collectively referred to by the program as CAV technologies. (FHWA, 2025)
- **Intent of the CAV Technology Tracking Resource:** To be an inventory that proactively identifies emerging CAV technology developments and capabilities, fleet composition, and current and future market penetration levels information, which can be used to inform more realistic AMS modeling assumptions and scenarios.

Mockup: CAV Technology Tracking Resource

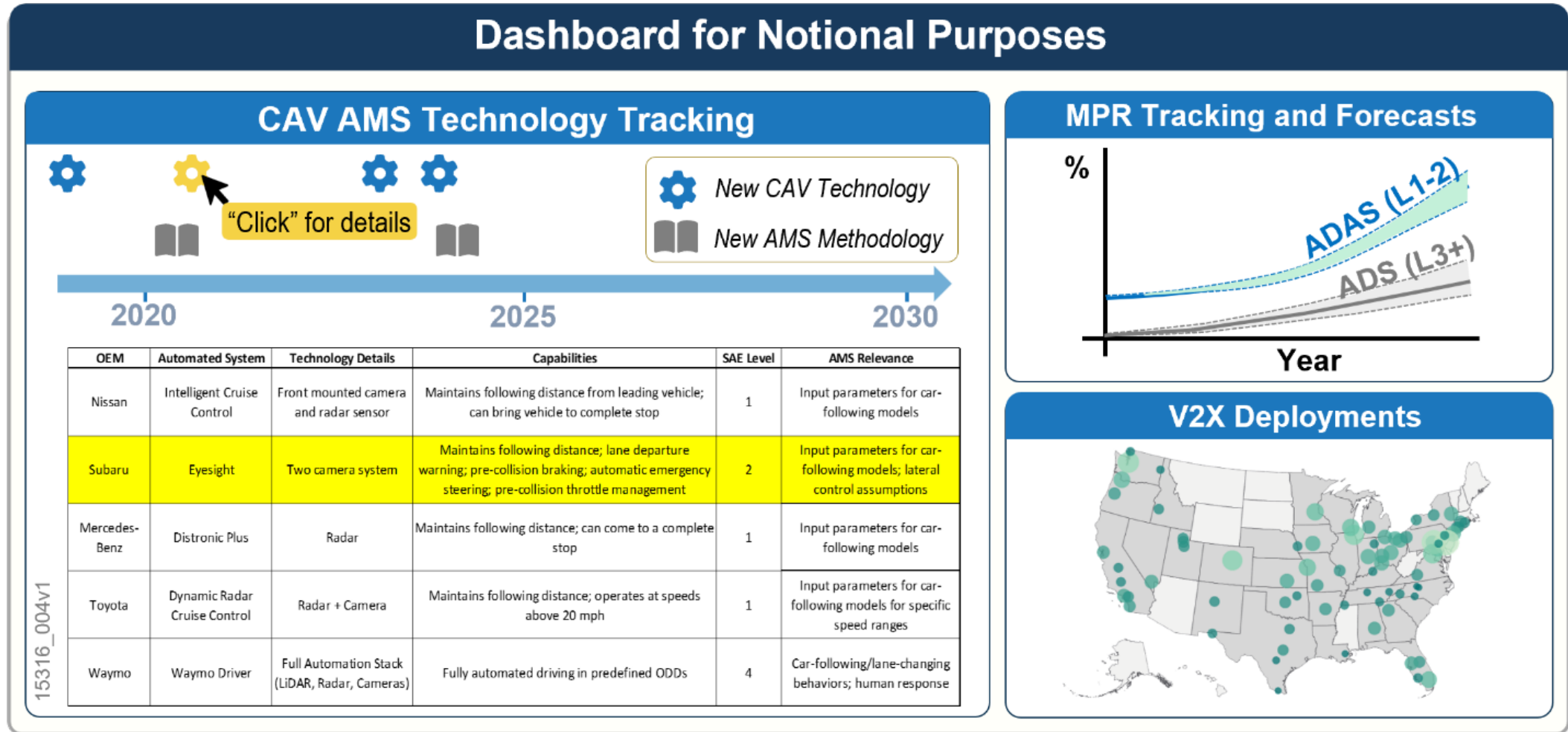


Image Source: Noblis

Motivation

- Current **CAV behaviors differ from human drivers** and are **not consistent with theoretical assumptions** (e.g., low response times, smaller headways)
- Field tests of commercial ACC showed:
 - Larger following distances when ACC was engaged (Makridis et al., 2020)
 - No significant improvement in response times compared to humans (Makridis et al., 2020)
 - Car following settings and performance vary by manufacturer (Li et al., 2021; Ma et al., 2025)
 - Faster deceleration and slower acceleration compared to human drivers (Metz, 2021)
 - Fewer lane changes when ACC was engaged (Metz, 2021)
 - String unstable (Gunter et al., 2020)

Key Takeaway: We need CAV data to accurately model emerging CAV technologies and subsequent driving behaviors (e.g., control algorithms, headway settings, lane-changing behaviors)

Interactive Feedback Activity for CAV Tracking Resource



CAV Technology/AMS Questions for Discussion

- What types of AMS tools do you/ your projects use?
- How do you consider or plan to consider CAVs in your future transportation plans?
- If you consider CAVs in your future plans, how do you currently represent emerging CAV technologies within the AMS tools you/ your projects?
 - What data/ sources do you use to justify specific CAV technology assumptions/ behavior/ performance?
 - Can these data/ sources be shared publicly?
- If you consider CAVs in your future plans, what are the most critical CAV data gaps for your specific AMS tasks?
- Which is more useful: (a) vehicle make/ model-level information (e.g., sensor types, software updates, CAV features, observed behaviors) or (b) aggregated fleet-level statistics (e.g., distribution of ACC following distances across manufacturers, penetration rates of ACC)?

Interactive Feedback/ Brainstorming Activity for CAV AMS Challenge Areas



General CAV AMS Challenge Areas for Discussion

- From the December 2025 ITS AMS CoP meeting, the following CV, AV, and CAV themes were identified:
 - **New data collection paradigms** focused on automated behavior rather than human behavior
 - **Communications modeling integration** with appropriate levels of complexity
 - **Guidance development** for practitioners at all planning horizons
 - **Model updates** from human-centric to automation-centric algorithms
 - **Safety analysis tools** that work in mixed traffic environments
 - **Standardization** across agencies, vendors, and OEMs
- Mural board activity to gain reactions, additional insights, etc. for these theme areas and initial insights from notes

References

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